Lecture 07:
Relational Data Model (Exercise)

ITCS216 Fundamentals of Database Systems

Semester 1/2026

Asst. Prof. Dr. Preecha Tangworakitthaworn preecha.tan@mahidol.ac.th
Asst. Prof. Dr. Siripen Pongpaichet siripen.pon@mahidol.ac.th
Asst. Prof. Dr. Jidapa Kraisangka jidapa.kra@mahidol.ac.th

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RECAP: Mapping ERD to Relational Model



8 Steps to Transform ERD to Relational Model

1. Transforming regular entity types
2. Transforming weak entities
3. Transforming binary 1:1 relationship types
4. Transforming binary 1:N relationship types
5. Transforming binary M:N relationship types
6. Transforming multi-valued attribute *A* in relation *K*
7. Transforming *n-ary* relationship types
8. Transforming specialization or generalization

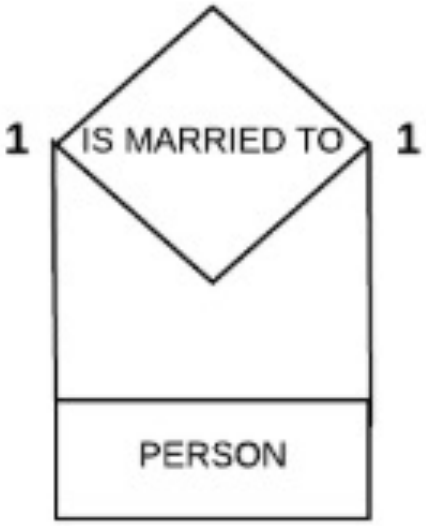
Each step can be skipped if your ERD/EERD does not contain those criteria

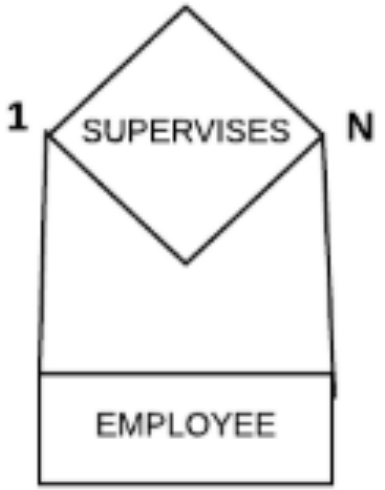
N-ary Relationship

- A relationship is an important part of any Entity relationship diagram as it shows the relation between two different entities.
- In an *n*-ary relationship, the *n* shows the number of entities in the relationship.
- It can be anything but the most popular relationships are unary, binary and ternary where the number of entities respectively are one, two and three.

Unary Relationship

When there is a relationship between two entities of the **same type**, it is known as a ***unary*** or ***recursive*** relationship.

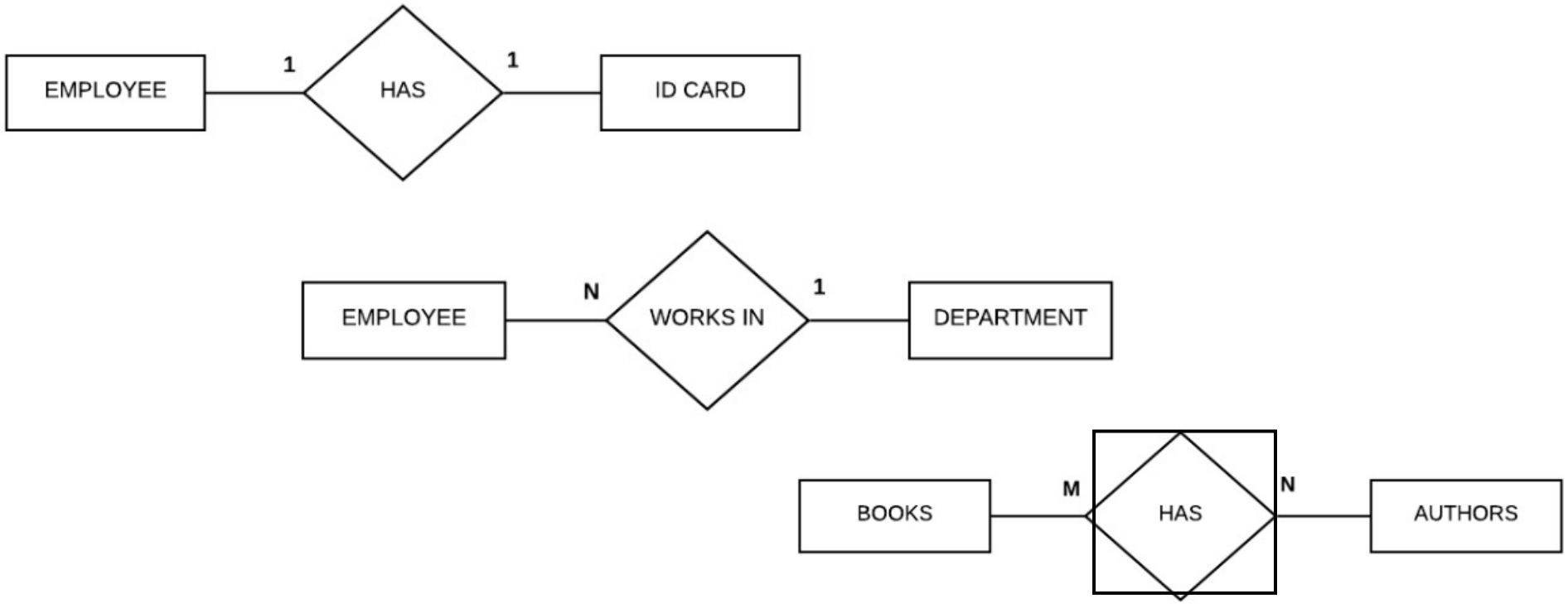




Ref: https://www.tutorialspoint.com/n-ary-relationship-in-database

Binary Relationship

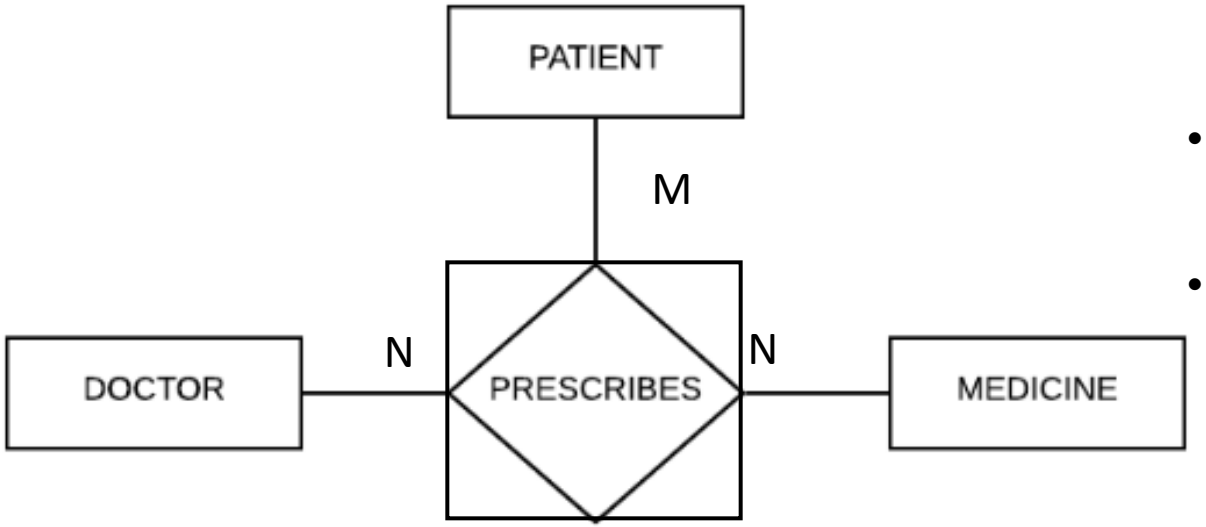
When there is a relationship between **two** different entities, it is known as a ***binary*** relationship.



Ref: https://www.tutorialspoint.com/n-ary-relationship-in-database

Ternary Relationship

When there is a relationship between **three** different entities, it is known as a ***ternary*** relationship.



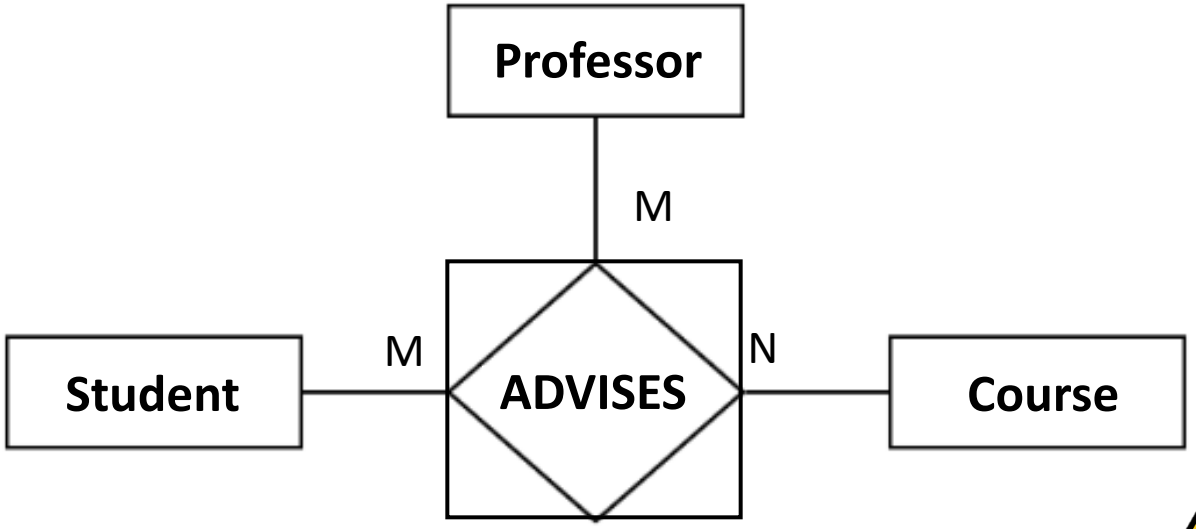
- A **DOCTOR** can prescribe many **MEDICINES** to many **PATIENTS**.
- A **PATIENT** can be prescribed many **MEDICINES** by many **DOCTORS**.
- A **MEDICINE** can be prescribed by many **DOCTORS** to many **PATIENTS**.

Ref: https://www.tutorialspoint.com/n-ary-relationship-in-database

What Does Each Number Mean?

- In a ternary relationship involving three entities (let's call them A, B, and C), the numbers (1 or M) tell you how many instances of a **single entity** can be associated with a **specific combination of instances from the other two entities**.
  - **1** means for a specific combination of A and B, there is **only one** instance of C.
  - **M** means for a specific combination of A and B, there can be **many** instances of C.
- The confusion often arises because the rule applies to all three combinations:
  - A-B pair and how it relates to C.
  - A-C pair and how it relates to B.
  - B-C pair and how it relates to A.

Ternary Relationship (cont.)



- Each **STUDENT-COURSE** pair can be advised by many **PROFESSORS (M)**
- Each **STUDENT-PROFESSOR** pair can be assigned for many different **COURSES (N)**
- Each **PROFESSOR-COURSE** pair can advise many **STUDENTS (M)**

These data are valid records for this relationship

Aj. Preecha – Advises – Mr. A – on Database Subject

Aj. Preecha – Advises – Mr. B – on Database Subject

Aj. Preecha – Advises – Mr. A – on Accounting

Aj. Jidapa – Advises – Mr. A – on Database Subject

Aj. Jidapa – Advises – Mr. B – on Programming

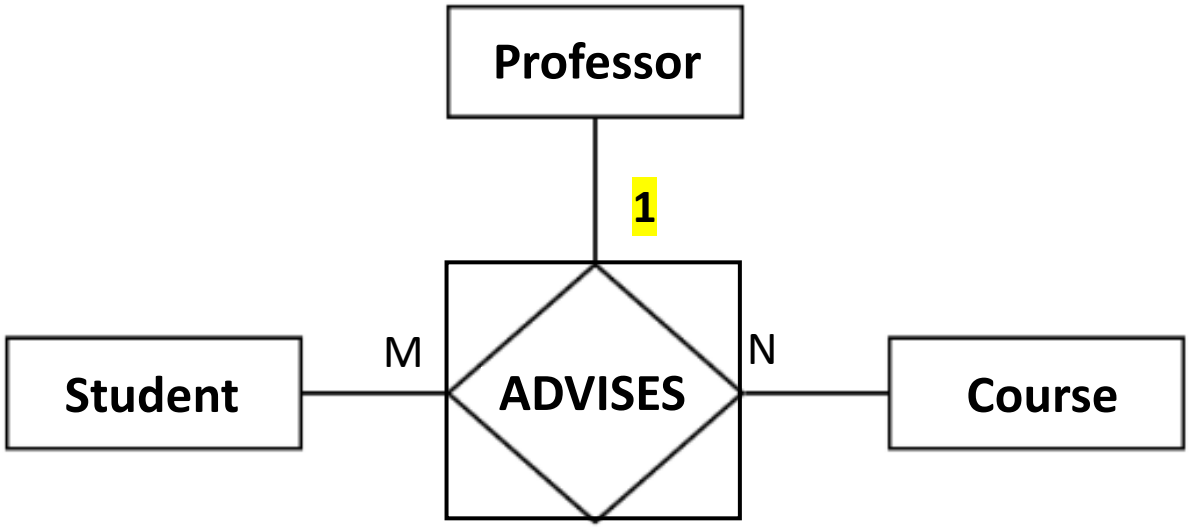
Aj. Siripen – Advises – Mr. A – on Database Subject

Problem in real world: Many professors can advise the same student on the same subject.

"Aj. Preecha, Aj. Jidapa, and Aj. Siripen advise Mr. A on Database Subject"

This may cause confusion and conflicting guidance.

Ternary Relationship (cont.)



- Each **STUDENT-COURSE** combination can be advised by only one **PROFESSOR (1)**
- Each **STUDENT-PROFESSOR** combination can be the designated advisor from many different **COURSES (N)**
- Each **PROFESSOR-COURSE** combination can advise many **STUDENTS (M)**

Some data are invalid for this relationship

Aj. Preecha – Advises – Mr. A – on Database Subject

Aj. Preecha – Advises – Mr. B – on Database Subject

Aj. Preecha – Advises – Mr. A – on Accounting

~~**Aj. Jidapa – Advises – Mr. A – on Database Subject**~~ **(Invalid)**

Aj. Jidapa – Advises – Mr. B – on Programming

~~**Aj. Siripen – Advises – Mr. A – on Database Subject**~~ **(Invalid)**



"For Database Subject, Mr. A can have only advisor (Aj. Preecha)"

One advisor per course per student to avoid confusion and conflicting guidance.

Identify the Need for a Ternary Relationship

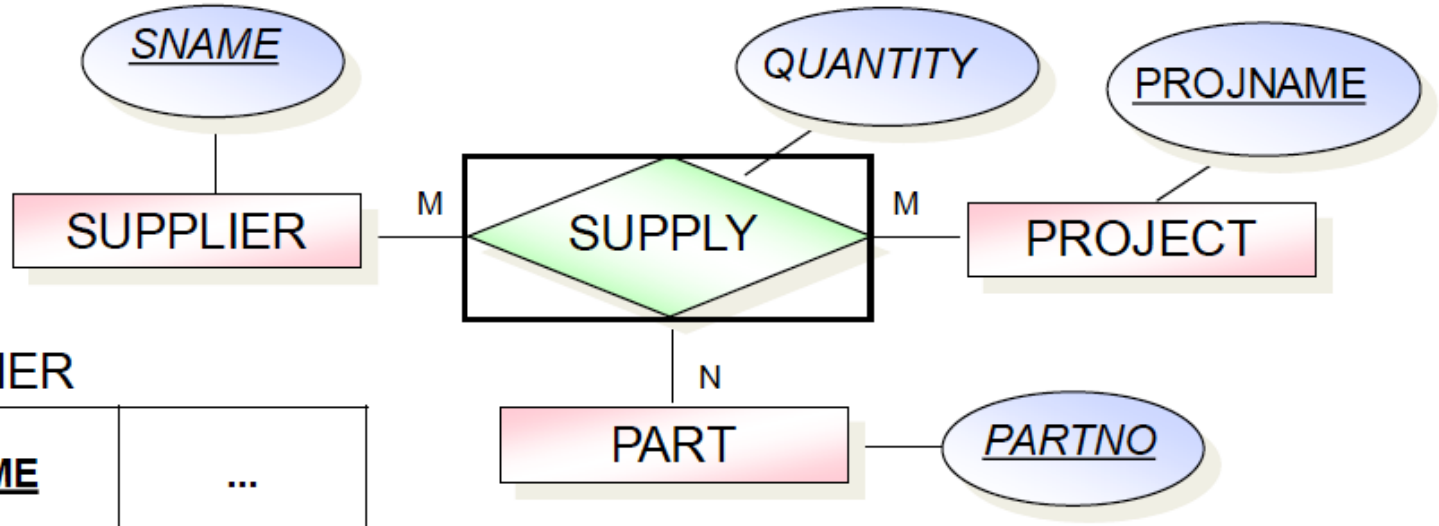
- **Check for Simultaneous Connection:**
  - A ternary relationship is necessary only when the association fundamentally requires all three entities at once.
- **Ask Key Questions:**
  - Can the meaning of the relationship be accurately described without all three entities being involved simultaneously?
  - If you consider the entities in pairs, do the relationships of those pairs cover the whole of the original three-way relationship?
  - If you cannot define the relationship with only binary connections (pairs), then a ternary relationship is required.

Step 7 : N-ary Relationship

For each n-ary relationship type R (n>2)

- Create a new relation **E** to represent R
- Include the FK in S obtained from the PK of the participating entities
- Include the attributes of the n-ary relationship to **E**

Step 7: N-ary Relationship



SUPPLIER

SNAME	...

Each supplier supplies several parts for any one of many projects. Each project has many parts from any one of many suppliers.

PROJECT

PROJNAME	...

PART

PARTNO	...

SUPPLY

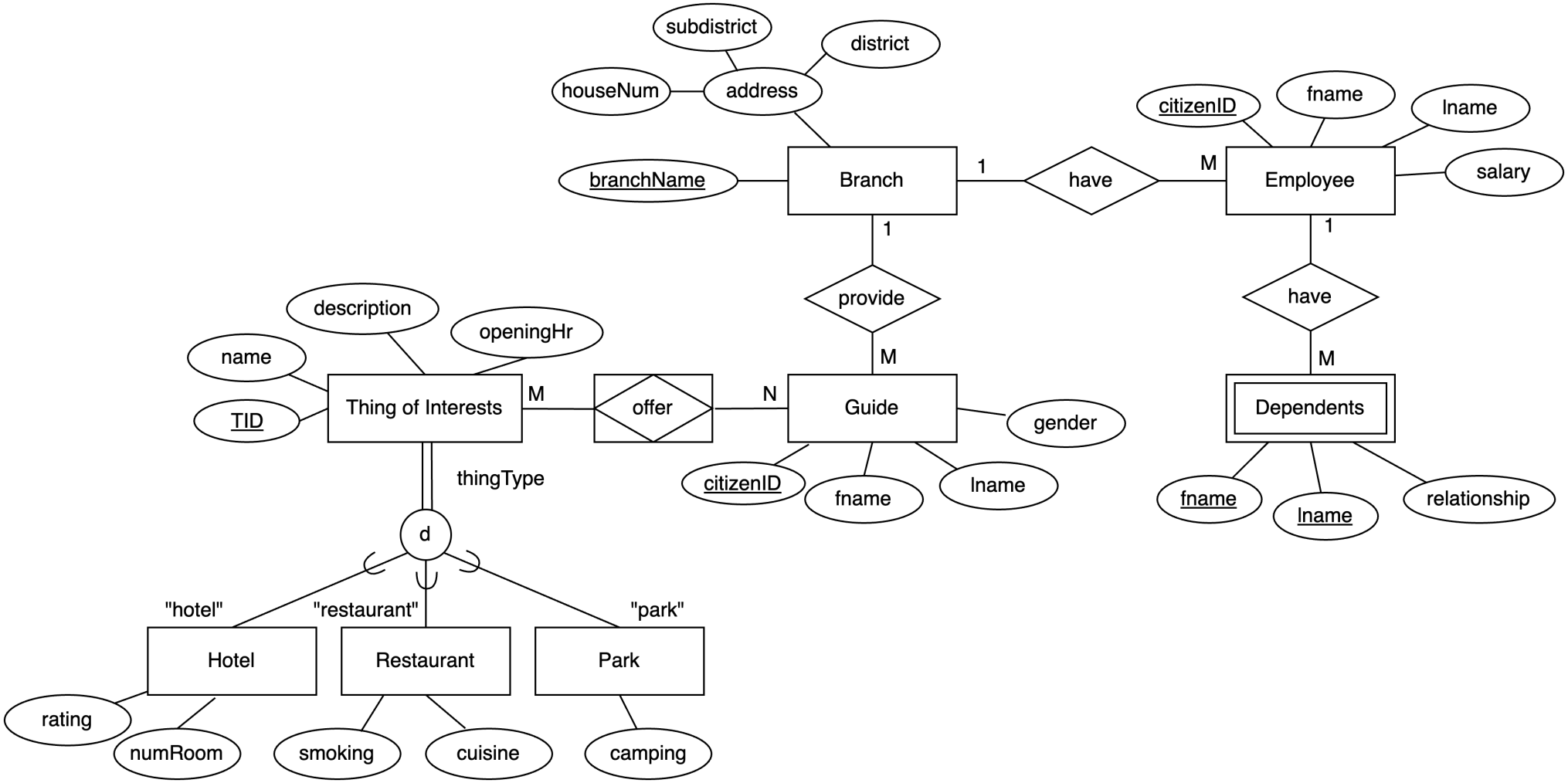
SNAME	PROJNAME	PARTNO	QUANTITY

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Revisit

Thai Travel Agency

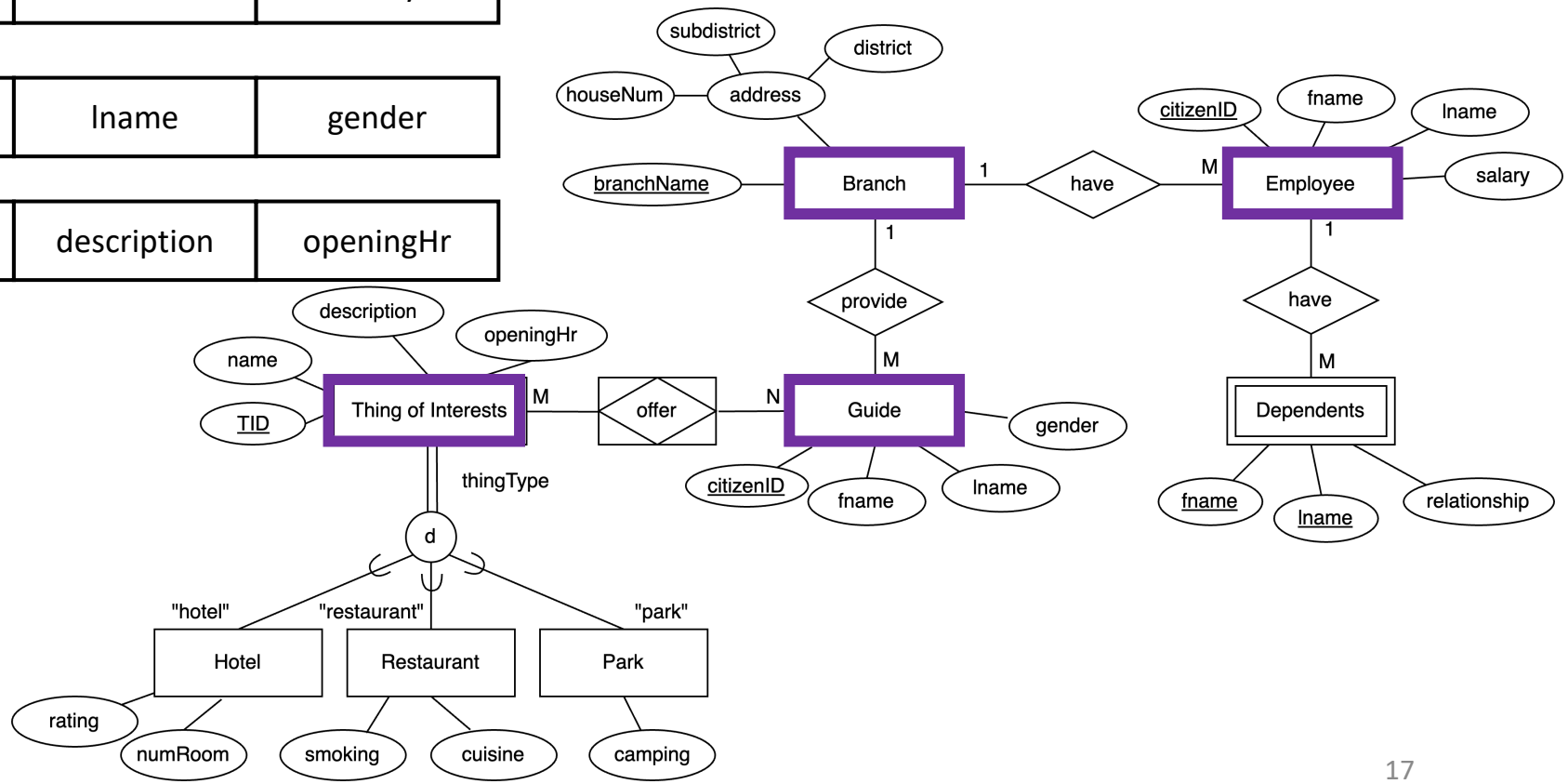
Thai Travel Agency



Step 1: Regualar Entity

address

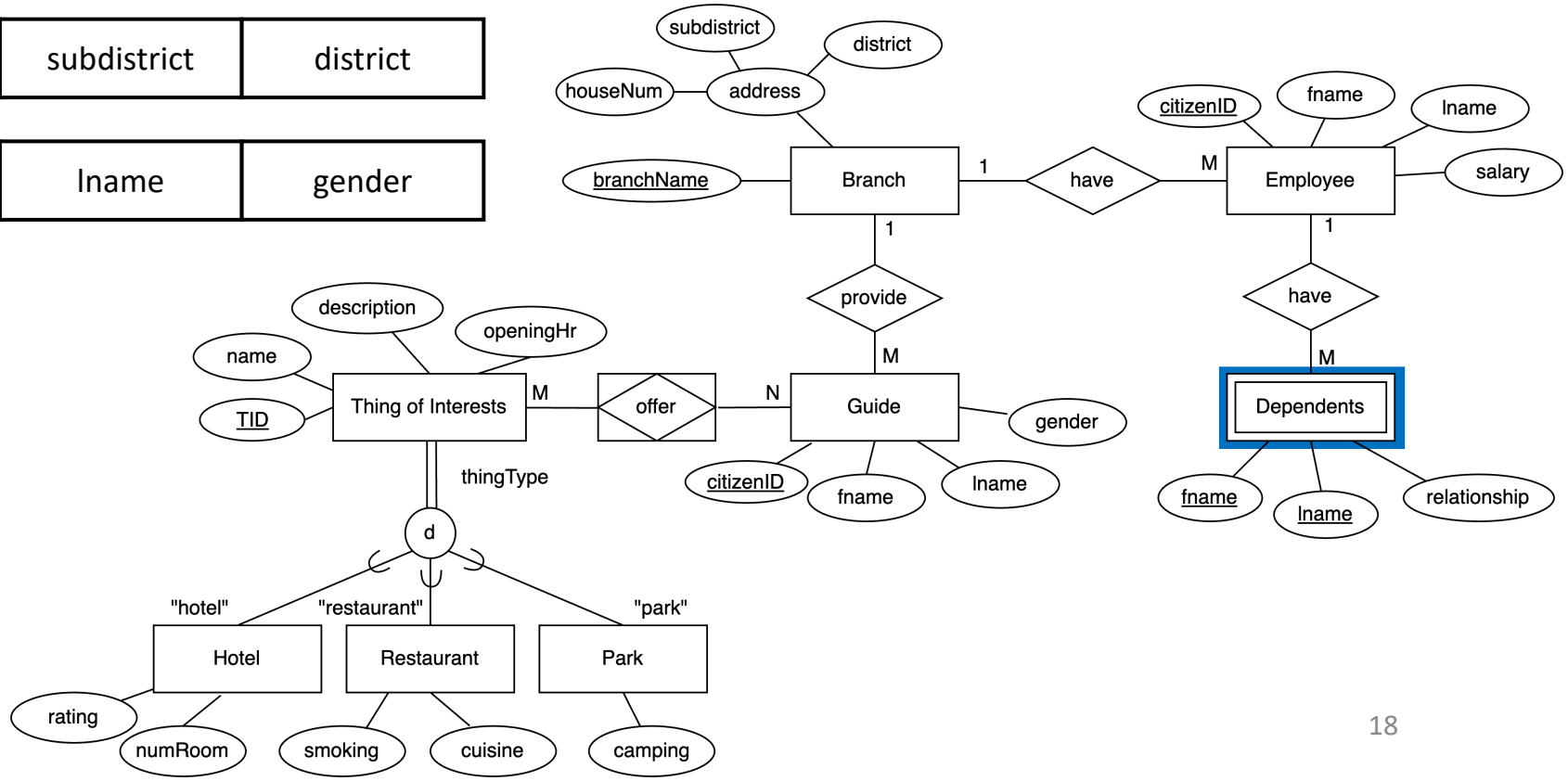
Branch	branchName	houseNum	subdistrict	district
Employee	ecitizenID	fname	lname	salary
Guide	gcitizenID	fname	lname	gender
TOI	TID	name	description	openingHr



Step 2: Weak Entity

FK PK

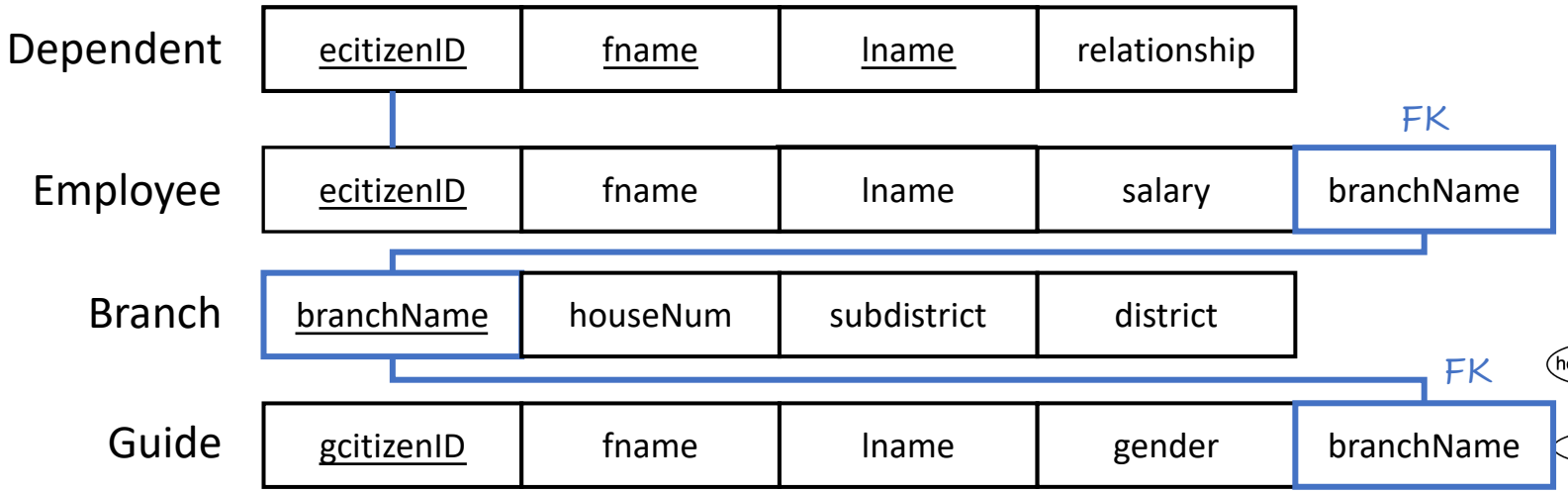
Dependents	ecitizenID	fname	lname	relationship
Employee	ecitizenID	fname	lname	salary
Branch	branchName	houseNum	subdistrict	district
Guide	gcitizenID	fname	lname	gender

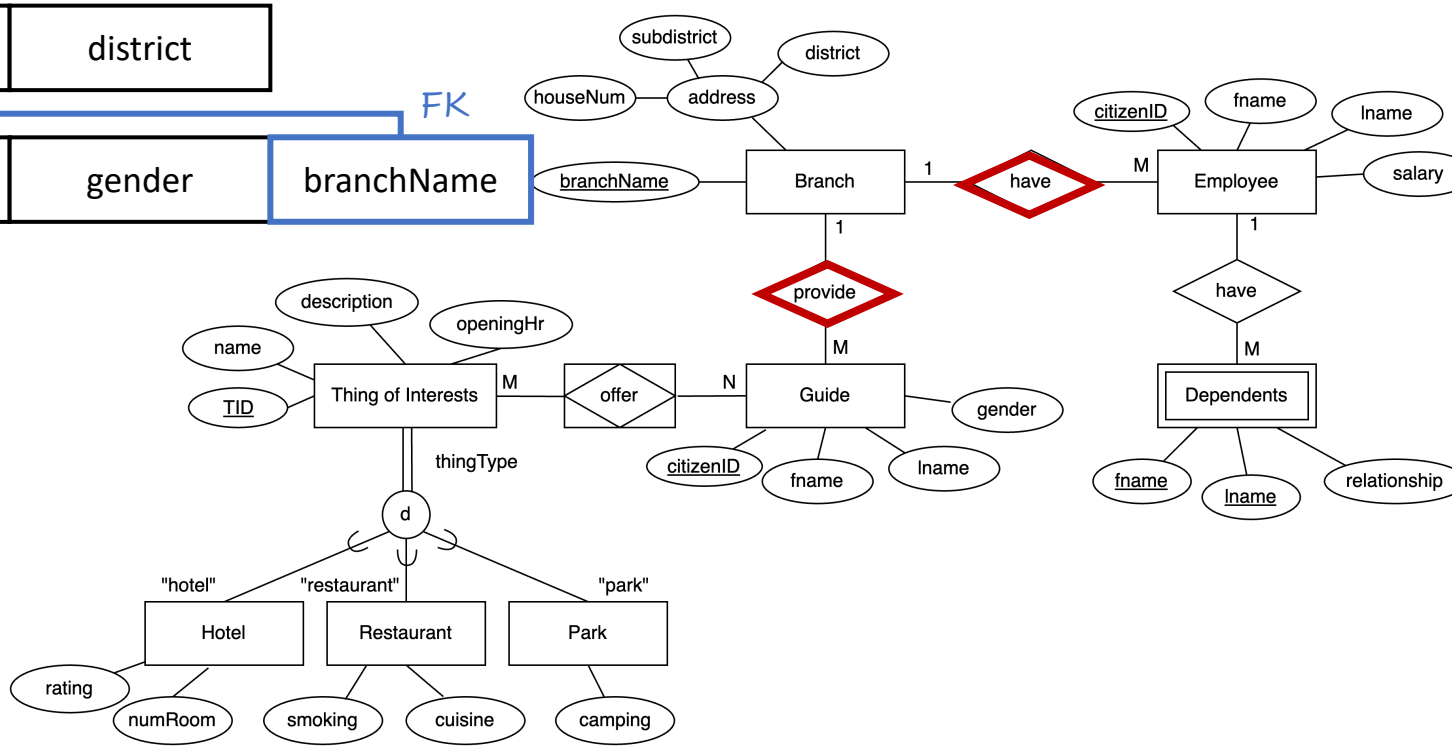


Step 3: 1-1 Relationship

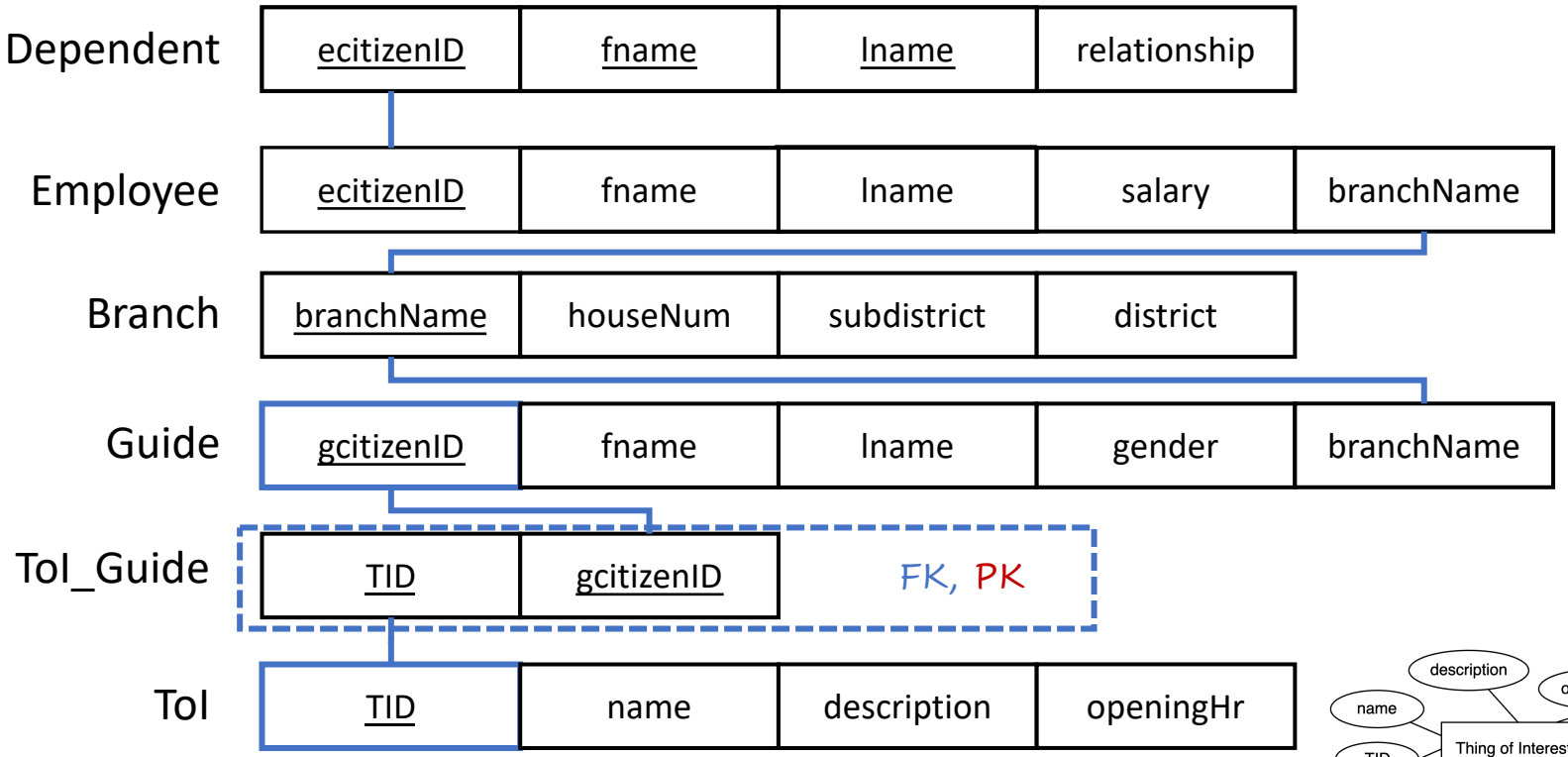
“No 1-1 Relationship”

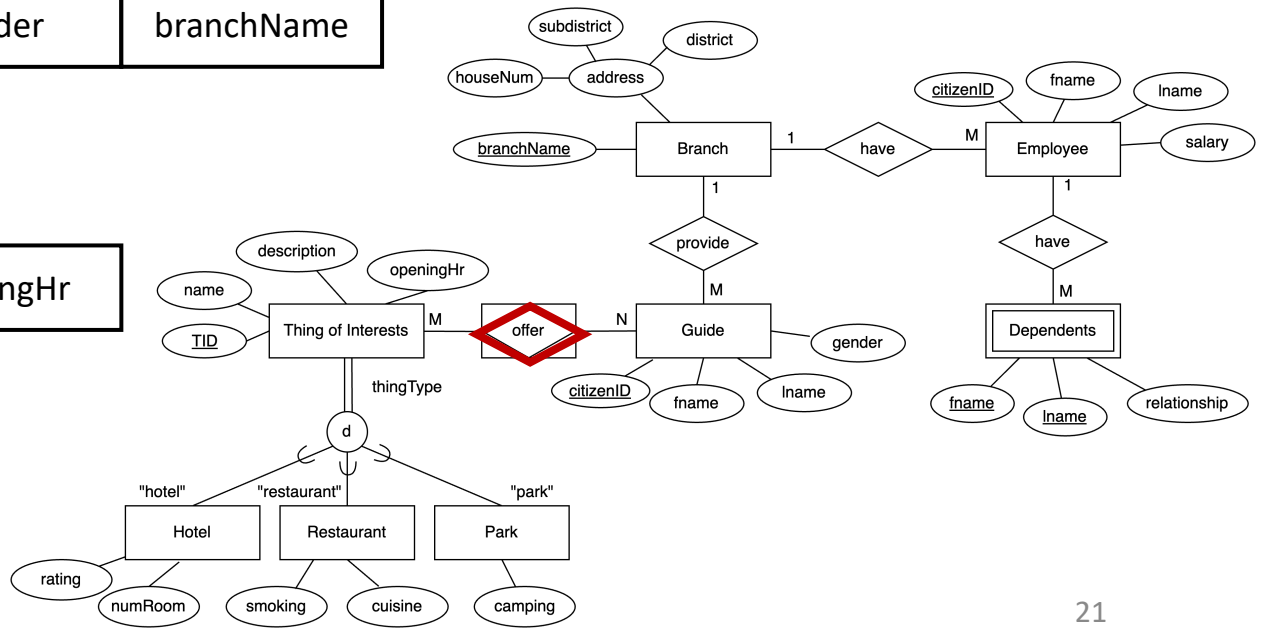
Step 4: 1-M Relationship





Step 5: M-N Relationship





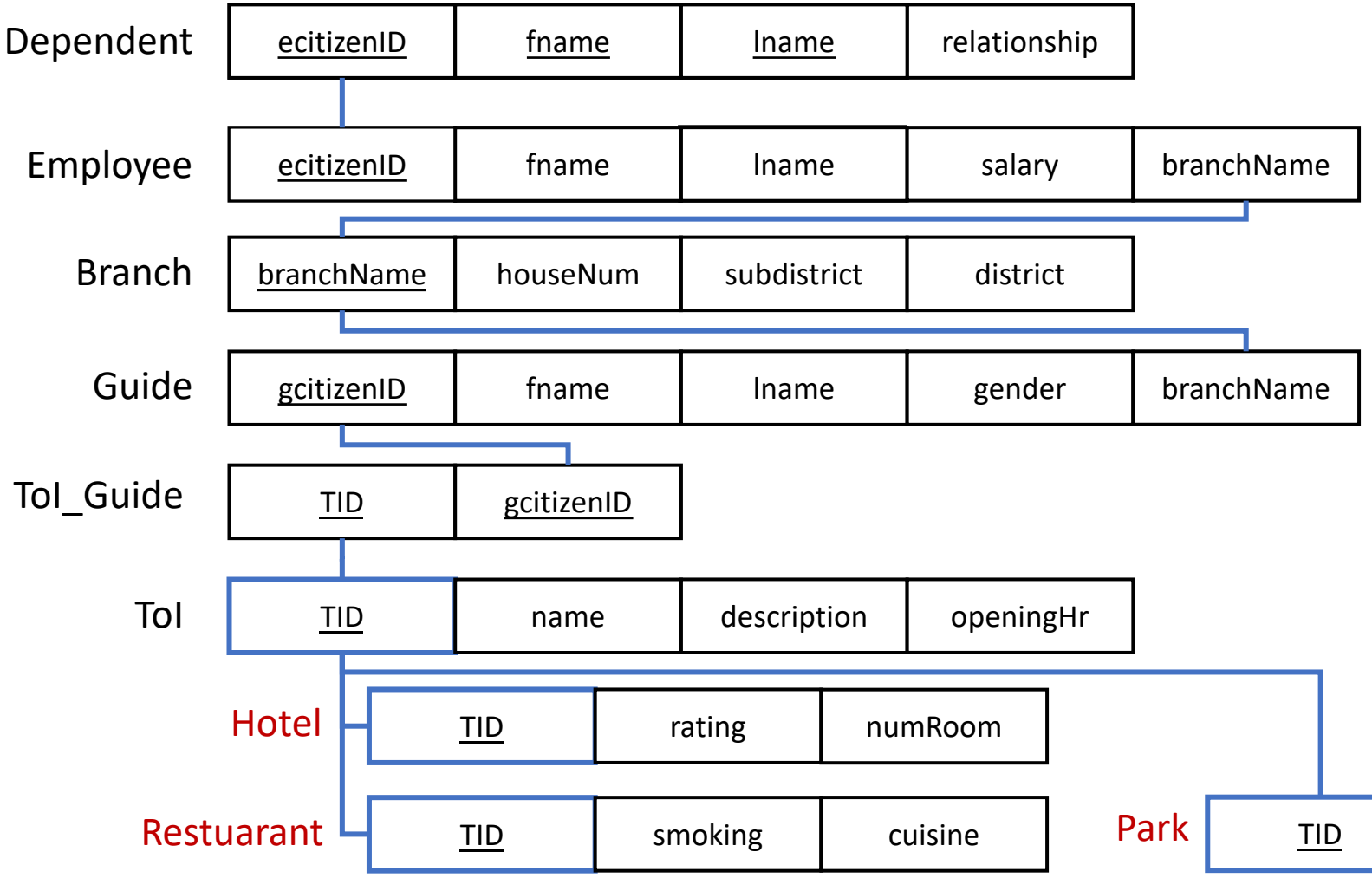
Step 6: Multi-valued Attributes

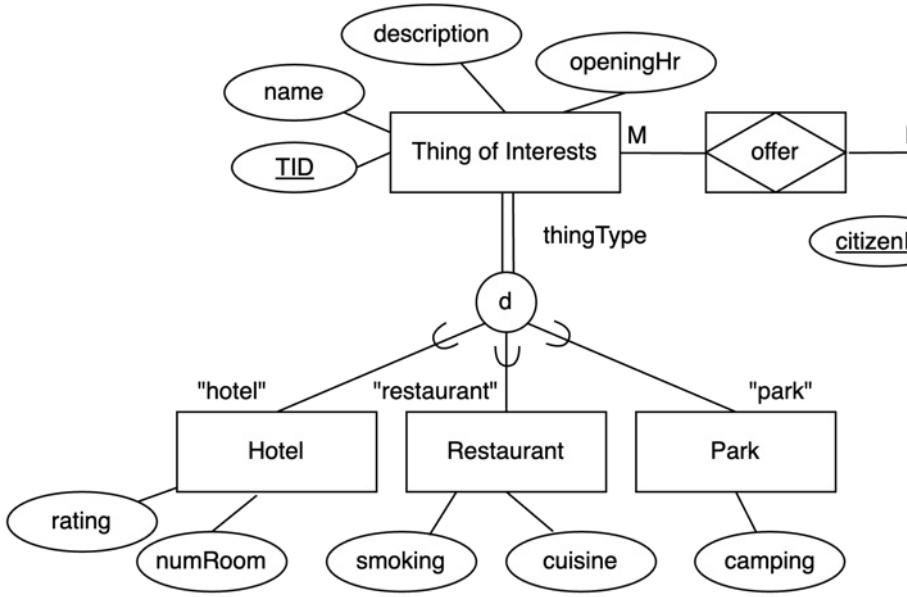
“No Multi-valued Attributes”

Step 7: N-ary Relationship

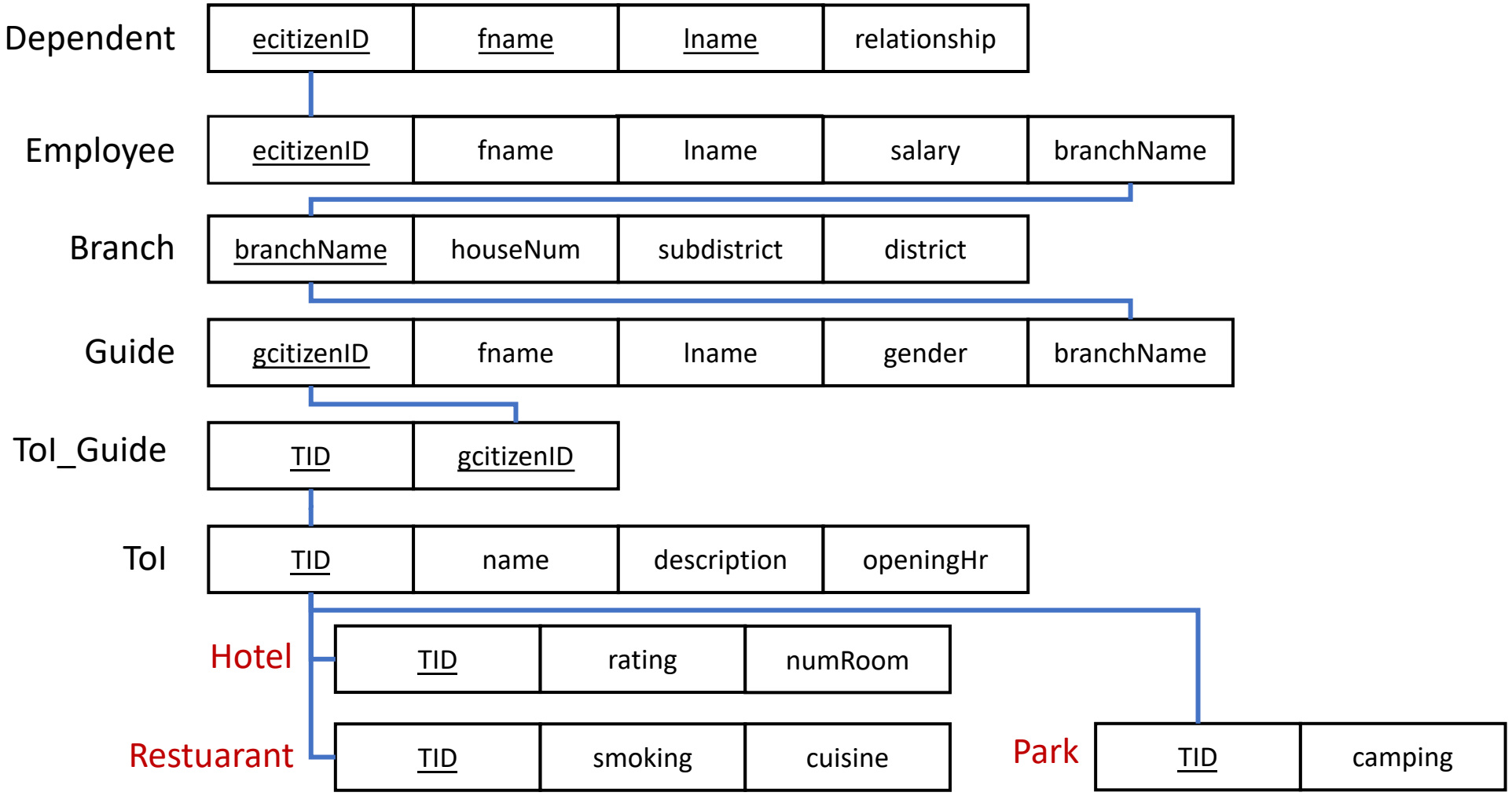
“No N-ary Relationship”

Step 8: Specialization (8A)





Final Relational Model



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Lecture 07:
Mapping Relational Model

* * 1.5 hr **

Midterm Exam Guideline

Midterm Exam Guideline

- A *3-hour* paper-based exam
- Bring your own stationary:
- A Limited-note Exam
  - A single sheet of A4 paper is allowed
  - Dictionary or any books are not allowed
- Exam: 25% of Your Grade
- Topic Distribution
  - DB Concept and Theory ~ 25% (**Hint**: Lecture 1 – Lecture 5)
  - Requirement Analysis ~ 25% (**Hint**: In-class exercise, Workshop, Project)
    - Business Rules
    - Business Constraint
  - EER + EERD ~ 50% (**Hint**: Lecture 3 - 5, Workshop, Project)